

Sphero Blueprint Snap Class Pack

WHAT'S IN THE KIT

- 16 Blueprint Snap kits with tool-free snap connectors, beams, gears, and motion pieces



QUICK-START GUIDE

1. Pieces snap together by hand — no tools needed.
2. Run a 10-minute free-build first so students learn the connections.
3. Follow a guided build, then remix it with a 'what if' twist.
4. Use the challenge cards/lessons from Sphero EDU.

FULL LESSONS IN THIS GUIDE

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PRO TIP: Snap and the full Blueprint system share DNA — Snap in elementary sets up Blueprint in secondary.

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LESSON · GRADES 3-5

Snap Machines

LEARNING OUTCOMES

- Students will build a machine that transfers motion
- Students will use push, pull, and turn vocabulary accurately
- Students will explain how parts work together in a system

MATERIALS

- Blueprint Snap kits (1 per pair)
- Force-and-motion word wall cards
- Target objects (erasers, pom-poms) for machines to move

INSTRUCTIONS

1. Free-build 10 minutes so hands learn the snap connections.
2. Class builds one guided machine (spinner or gripper) together step-by-step.
3. Challenge: modify or rebuild so your machine moves a pom-pom without hands touching it.
4. Machine tour: each pair demonstrates and narrates using push/pull/turn words.
5. Word-wall check: tag each machine with the forces it uses.
6. Reset ritual: pieces sorted back by shape into the kit.

ACTIVITY

Snap Machines — build a working machine, then explain which parts push, pull, or turn. The no-hands pom-pom challenge forces genuine mechanism thinking at elementary speed.

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LESSON · GRADES 6-8

Invention Iteration Sprint

LEARNING OUTCOMES

- Students will iterate a design under constraints across timed rounds
- Students will document design changes and their effects
- Students will evaluate which iteration strategies produced improvement

MATERIALS

- Blueprint Snap kits (1 per team)
- Constraint card: 'lift this eraser 10 cm' (or similar)
- Iteration journal (one entry per round)
- Timer

INSTRUCTIONS

1. Reveal the constraint and give 3 minutes of sketch-planning, no pieces.
2. Round 1 (8 min): build a first attempt; test; journal what worked and what didn't.
3. Round 2 (8 min): improve — journal the specific change made and why.
4. Round 3 (6 min): final refinement round with the same journaling.
5. Score on function AND on journal quality — evidence of deliberate change.
6. Debrief: which teams changed one thing at a time, and did it pay off?

ACTIVITY

Invention Iteration Sprint — three timed build rounds with mandatory change-logging between each. The journal grade makes iteration the assignment, not an accident.